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Aim: Data Visualization with Matplotlib: Data visualization on a dataset using matplotlib and seaborn libraries

• Scatter plot • Line plot • Bar plot • Histogram • Box plot • Pair plot

What is the importance of data visualization in data analysis? How does visualization help in exploratory data analysis (EDA)? ::Importance:

Data visualization helps present data in a visual format (graphs, charts). It makes complex data easy to understand. Helps in quick decision-making. Identifies patterns, trends, and relationships in data. Useful for communicating insights effectively.

Role in Exploratory Data Analysis (EDA):

Helps detect outliers and anomalies. Shows distribution of data (using histograms, box plots). Helps understand relationships between variables (scatter plot, pair plot). Assists in feature selection for modeling.

What is the difference between Matplotlib and other visualization libraries? Matplotlib: Basic and most widely used plotting library. Provides full control over plots. Good for simple and customized graphs.

Other Libraries (like Seaborn): Built on top of Matplotlib. Provides better design and attractive visuals. Easier for statistical visualization. Requires less code for complex plots.

What is the purpose of the pyplot module in Matplotlib? pyplot is a module in Matplotlib used to: Create plots easily using simple functions. Acts like MATLAB-style interface. Used for: Creating figures Plotting graphs Adding labels, titles, legends

Example:

```
import matplotlib.pyplot as plt
plt.plot(x, y)
plt.show()
```

Explain the process of creating a basic line plot. Steps:

1.Import library:

```
import matplotlib.pyplot as plt
```

2.Define data:

```
x = [1, 2, 3, 4]
y = [10, 20, 25, 30]
```

3.Create plot:

```
plt.plot(x, y)
```

4.Add labels and title:

```
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Basic Line Plot")
```

5.Display plot:

```
plt.show()
```

Why is customization important in data visualization? What customization options are available in Matplotlib? Why Customization is Important:

Makes graphs more attractive and readable. Helps highlight important information. Improves presentation quality. Makes data easier to interpret.

Customization Options in Matplotlib:

Colors (color='red') Line styles (linestyle='--') Markers (marker='o') Titles and labels Legends Grid lines
Figure size Font size and style Axis limits

Example:

```
plt.plot(x, y, color='blue', linestyle='--', marker='o') plt.title("Customized Plot") plt.grid(True)
```

Conclusion

Data visualization is an essential part of data analysis as it helps in understanding complex data easily through graphical representation. In this assignment, various plots such as scatter plot, line plot, bar plot, histogram, box plot, and pair plot were created using Matplotlib and Seaborn. These visualizations help in identifying trends, patterns, relationships, and distributions within the dataset. Matplotlib provides flexibility and customization, while Seaborn enhances the visual appeal and simplifies complex plotting. Overall, data visualization plays a crucial role in exploratory data analysis and supports better decision-making.

```
# 1. Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 2. Basic Line Plot Example (Matplotlib)
x_simple = [1, 2, 3, 4]
y_simple = [10, 20, 25, 30]

plt.figure(figsize=(8, 5))
plt.plot(x_simple, y_simple, color='blue', linestyle='--', marker='o', label='Data Line')
plt.title("Customized Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.grid(True)
plt.legend()
plt.show()

# 3. Data Visualization on Tips Dataset
# Note: Using Seaborn's built-in tips dataset for compatibility
df = sns.load_dataset('tips')

# Check first few rows
print("Dataset Head:")
print(df.head())

# 4. Scatter Plot: Total Bill vs Tip
plt.figure(figsize=(10, 6))
sns.scatterplot(x='total_bill', y='tip', hue='sex', data=df)
plt.title("Scatter Plot: Total Bill vs Tip")
plt.xlabel("Total Bill")
plt.ylabel("Tip")
plt.show()

# 5. Line Plot: Total Bill Trend
plt.figure(figsize=(10, 6))
sns.lineplot(x=df.index, y='total_bill', data=df)
plt.title("Line Plot: Total Bill Trend")
plt.xlabel("Index")
plt.ylabel("Total Bill")
```

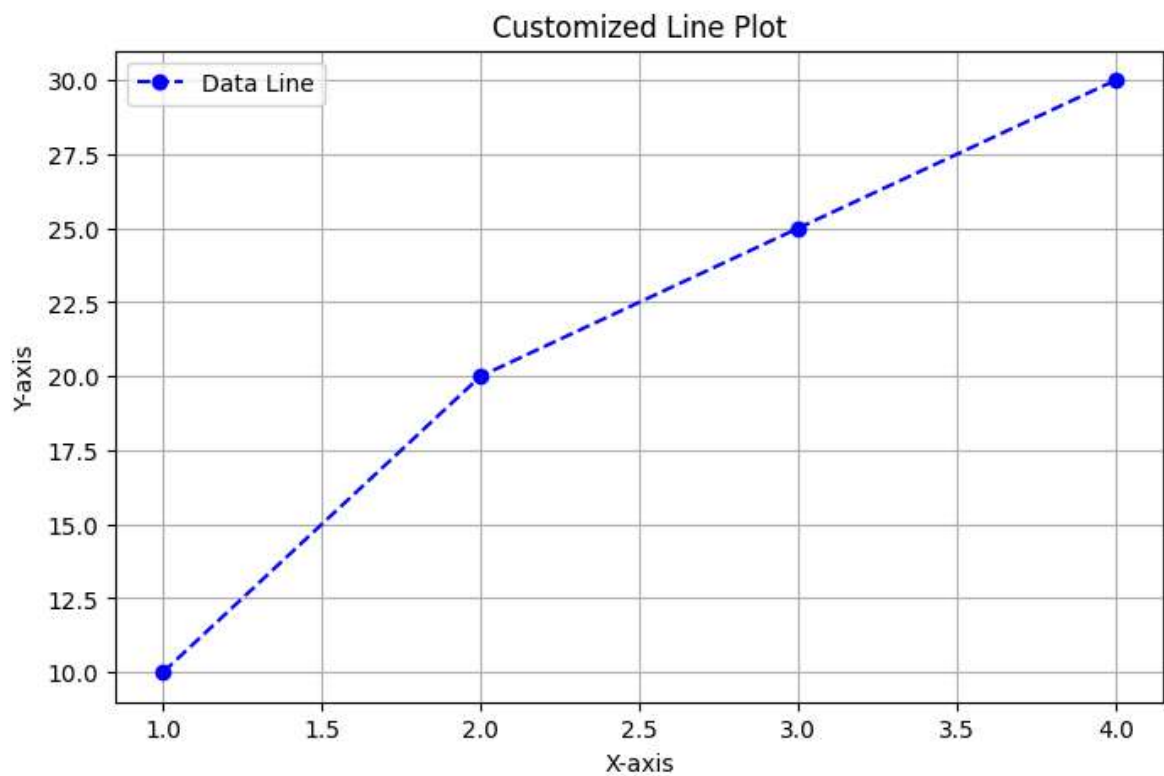
```
plt.show()

# 6. Bar Plot: Average Bill per Day
plt.figure(figsize=(10, 6))
sns.barplot(x='day', y='total_bill', data=df)
plt.title("Bar Plot: Average Bill per Day")
plt.xlabel("Day")
plt.ylabel("Average Total Bill")
plt.show()

# 7. Histogram: Total Bill Distribution
plt.figure(figsize=(10, 6))
sns.histplot(df['total_bill'], kde=True)
plt.title("Histogram: Total Bill Distribution")
plt.xlabel("Total Bill")
plt.ylabel("Frequency")
plt.show()

# 8. Box Plot: Tip by Gender
plt.figure(figsize=(10, 6))
sns.boxplot(x='sex', y='tip', data=df)
plt.title("Box Plot: Tip by Gender")
plt.xlabel("Gender")
plt.ylabel("Tip")
plt.show()

# 9. Pair Plot
print("Generating Pair Plot... (this may take a moment)")
sns.pairplot(df, hue='sex')
plt.show()
```

Dataset Head:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

